carb_coefs.cbn (deprecated)

`carb_coefs.cbn` is no longer read.

> **Removed**
>
> SWAT+ 62.0.1 ignores a copy of `carb_coefs.cbn` left in the run directory. Its settings now come from `carbon.bsn` and `carbon_lyr.bsn`.

Replacement files

The dynamic carbon model is switched on with `carbon` = 2 in `codes.bsn`. In earlier revisions the value was 1; in 62.0.1 the value 1 is reserved for the C-FARM model, which is not implemented. With `carbon` = 0 soil carbon is static and neither replacement file is read.

With `carbon` = 2 SWAT+ needs two files, both with a title line, a header line and data rows like other SWAT+ inputs:

- `carbon.bsn`: one row of 28 basin-wide values. It is named on the basin line of `file.cio` (default `carbon.bsn`).
- `carbon_lyr.bsn`: decomposition rates and CO2 fractions, one row for the top soil layer (`layer` 1) and one for the layers below (`layer` 2). Its name is built from the `carbon.bsn` name by adding `_lyr` before the extension, so it is not listed in `file.cio`.

If either file is missing, or `carbon.bsn` does not have 28 values, SWAT+ stops with an error. There is no fallback to built-in values.

An optional file `carbon_layers.prt` sets how many soil layers the per-layer carbon outputs include. Without it, the outputs cover the largest number of layers of any HRU.

Keyword migration

Keyword in `carb_coefs.cbn`	Replacement
`cbn_diagnostics`	Removed. Carbon outputs are switched on per file in `print.prt` with the `hru_cb_*` and `lsu_cb_*` rows.
`hp_rate`, `hs_rate`, `microb_rate`, `meta_rate`, `str_rate`, `microb_top_rate`, `hs_hp`	Columns of the same names in `carbon_lyr.bsn`.
`a1co2`, `asco2`, `apco2`, `abco2`	Columns of the same names in `carbon_lyr.bsn`.
`org_frac` (4 values)	`init_seq`, `init_microb`, `init_slow`, `init_passive` in `carbon.bsn`.
`prmt_21`	`koc_c` in `carbon.bsn`.
`prmt_44`	`solc_ratio` in `carbon.bsn`.
`till_eff_days`	`till_eff_days` in `carbon.bsn`.
`rtof`	`manure_c_frac` in `carbon.bsn`.
`cbn_consolidation_factors` (2 values)	`bio_consol`, `till_consol` in `carbon.bsn`.
`cbn_factor_approaches` (2 values)	`tmpf_eqn`, `watf_eqn` in `carbon.bsn`.
`tn`, `top`, `tx`	`t_cbn_min`, `t_cbn_opt`, `t_cbn_max` in `carbon.bsn`.
`zz_bmix_coefs` (3 values)	`bmix_a`, `bmix_b`, `bmix_c` in `carbon.bsn`.
`zz_emix_coefs` (3 values)	`tillmix_a`, `tillmix_b`, `tillmix_c` in `carbon.bsn`.
`photo_degrade_factor`	`sfc_rsd_photodeg` in `carbon.bsn`.
`nmbr_soil_test_layers`, `soil_test`	Removed, no replacement.

Six values that were fixed in the code are now columns of `carbon.bsn`: `n_act_frac` (fraction of humus nitrogen kept in the active pool), `cnr_cap` and `cnr_ref` (cap and reference C:N ratio of surface residue, formerly 500 and

25), `cpr_cap` and `cpr_ref` (cap and reference C:P ratio, formerly 5000 and 200), and `mathers_method` (method for the initial slow humus pool).

Example

`refdata/Ames_sub1` in the SWAT+ repository has both replacement files, converted from its former `carb_coefs.cbn`:

`carbon.bsn: derived from carb_coefs.cbn (basin carbon scalars)`

init_seq	init_microb	init_slow	init_passive	koc_c	solc_rat
0.95	0.02	0.44	0.54	1000.0	0

`carbon_lyr.bsn: derived from carb_coefs.cbn (per-layer carbon coefficients)`

layer	hp_rate	hs_rate	microb_rate	meta_rate	str_rate	microb_to
1	1.2e-05	2.92e-04	0.0164	0.0405	0.0107	
2	1.2e-05	1.81e-04	0.02	0.0507	0.0134	

Related files

- carbon.bsn and carbon_lyr.bsn: the replacement inputs.
- codes.bsn: the `carbon` switch.
- print.prt: carbon output switches.
- Carbon outputs: the files written by the dynamic carbon model.

Sources

- Readers: carbon_bsn_read.f90 (reads `carbon.bsn` and `carbon_lyr.bsn`), carbon_layers_read.f90, both called from proc_bsn.f90
- Variables: carbon_module.f90
- Carbon switch: `bsn_cc%cswat` in basin_module.f90
- Output switches: basin_print_codes_read.f90
- File name slot: `in_basin%carbon_bsn` in input_file_module.f90